Price Elasticity of Demand

for Health Care

What is Price Elasticity of Demand?

PED = % Change in Quantity Demanded ÷ % Change in Price

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Measures sensitivity

How much does quantity demanded change when price changes?

➗

It is a ratio

Always expressed as a coefficient — a pure number with no units.

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Interpreted by magnitude

$|PED| > 1 \rightarrow$ elastic; $|PED| < 1 \rightarrow$ inelastic; $|PED| = 1 \rightarrow$ unit elastic.

Computing PED: Worked Examples

Example 1 — GP Visits (Inelastic)

A government hike raises the GP consultation fee from ₹100 to ₹120 (+20%). Monthly visits per household fall from 2.5 to 2.3.

```
% ΔQd = (2.3 − 2.5) / 2.5 × 100 = −8%
% ΔP  = (120 − 100) / 100 × 100 = +20%
PED = −8% ÷ 20% = −0.4
```

|PED| = 0.4 < 1 ⟶ Inelastic

Example 2 — Elective Dental Care (Elastic)

A dental clinic raises teeth-whitening fees from ₹3,000 to ₹3,600 (+20%). Bookings drop from 80 to 56 per month.

```
% ΔQd = (56 − 80) / 80 × 100 = −30%
% ΔP  = (3,600 − 3,000) / 3,000 × 100 = +20%
PED = −30% ÷ 20% = −1.5
```

|PED| = 1.5 > 1 ⟶ Elastic

Example 3 — Insulin (Near-Zero Elasticity)

Insulin price rises from ₹500 to ₹650 per vial (+30%). Monthly purchases by diabetic patients fall from 4.0 to 3.9 units.

```
% ΔQd = (3.9 − 4.0) / 4.0 × 100 = −2.5%
% ΔP  = (650 − 500) / 500 × 100 = +30%
PED = −2.5% ÷ 30% ≈ −0.08
```

|PED| ≈ 0.08 ⟶ Highly inelastic

Key Evidence: The RAND HIE

RAND Health Insurance Experiment · USA, 1971–82

-0.2	-0.1	-0.17
Overall PED (all care)	Inpatient care	Outpatient care

What does −0.2 mean?

A 10% rise in out-of-pocket costs → ≈ 2% fall in quantity demanded. Demand is inelastic — people do reduce care when it costs more, but not sharply. This has big implications for copay design in health insurance.